

Technical Threat Intelligence Analysis of Triton-related Intrusion Campaign (C0032)

- MITRE ATT&CK Campaign and Software Intelligence Report
 - Prepared By Intern ID: 2132
 - Topic: Campaign C0032 and Software S0320 – S0330
-

Introduction

Cyber Threat Intelligence (CTI) plays an important role in identifying, analyzing, and mitigating cyber threats targeting organizations and critical infrastructure. This report presents a technical analysis of the MITRE ATT&CK campaign C0032 and assigned software entries S0320–S0330.

The report focuses on:

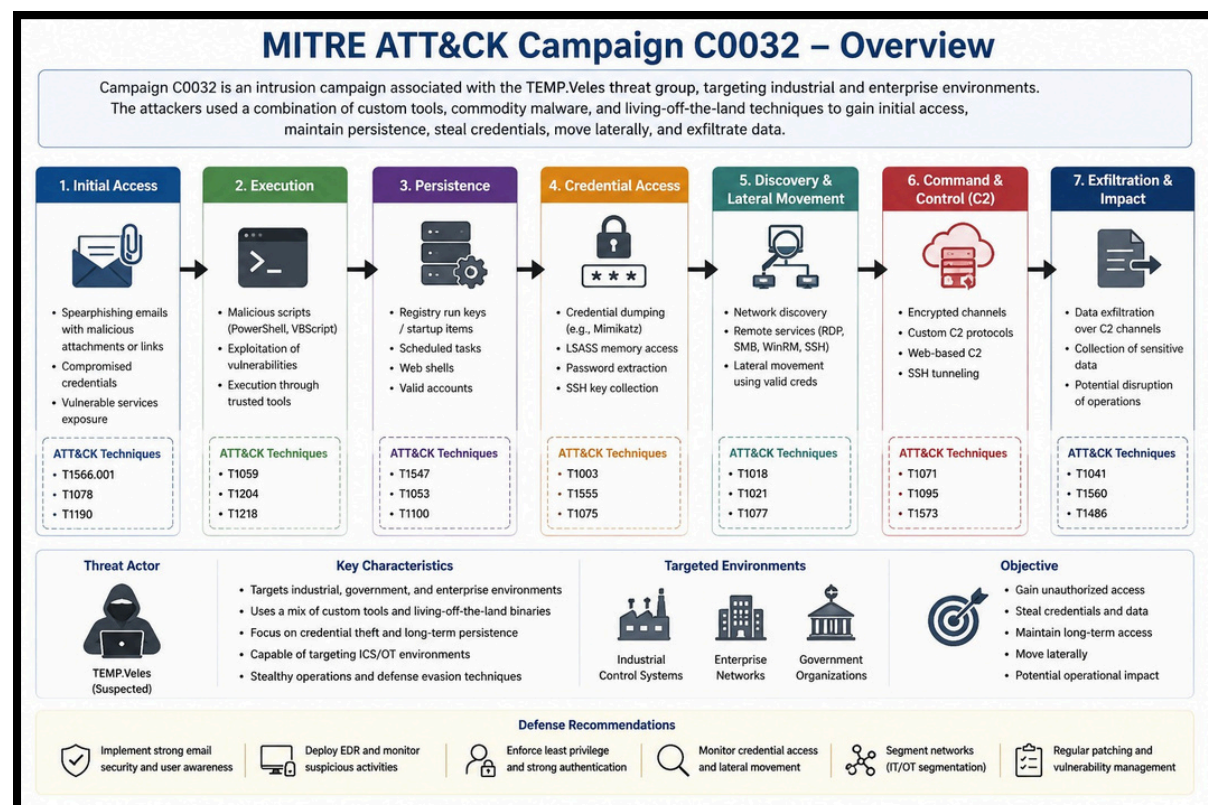
- Threat actor behavior and attack methodology
- Credential theft, persistence, and lateral movement
- MITRE ATT&CK technique mapping
- Malware and remote access tool analysis
- Indicators of Compromise (IoCs)
- Detection opportunities and mitigation strategies

The objective of this report is to demonstrate practical cyber threat [intelligence research](#), [malware](#) analysis, attack lifecycle understanding, and ATT&CK framework mapping through real-world cyberattack investigation and technical documentation.

Campaign Overview (C0032) :-

C0032 is a cyber intrusion campaign targeting enterprise

and industrial environments using advanced techniques such as credential theft, persistence, lateral movement, and remote access. The campaign demonstrates real-world attacker behavior and multi-stage cyberattack operations mapped to the MITRE ATT&CK framework.



Primary Objectives

- Unauthorized system access
- Persistence within compromised networks
- Credential compromise
- Lateral movement
- Data collection and exfiltration

Objectives of Attack

- Gain unauthorized access to networks
- Steal credentials and sensitive information
- Maintain persistence within systems
- Perform lateral movement across environments

- Exfiltrate collected data through secure channels
-

Threat Actor Profile – TEMP.Veles

- Threat Group - TEMP.Veles (also known as XENOTIME)
 - Threat Actor Type - State-sponsored industrial threat actor
 - **Characteristics :-**
 - Advanced operational capabilities
 - Focus on industrial infrastructure
 - Uses customized malware
 - Conducts stealthy lateral movement
 - Employs credential theft and remote access techniques
 - **Known Activities :-**
 - ICS targeting
 - SIS manipulation
 - Malware deployment
 - Credential harvesting
 - Network reconnaissance
 - **Associated Malware :-**
 - Triton / TRISIS / HatMan
 - Cobalt Strike
 - PowerShell payloads
-

Attack Timeline :-

Year	Attack Timeline	Description
2014 – 2016	Initial Preparation	Threat actors developed infrastructure,

		reconnaissance capabilities, and Triton malware targeting industrial environments.
--	--	--

Early 2017	Initial Intrusion	Attackers gained unauthorized access to the targeted industrial network using remote access methods and compromised credentials.
------------	-------------------	--

Mid 2017	Internal Reconnaissance	The attackers explored the network, identified operational technology systems, and collected information about SIS controllers.
----------	-------------------------	---

Mid 2017	Persistence & Lateral Movement	Malicious tools and remote administration utilities were deployed to maintain
----------	--------------------------------	---

		persistence and move across systems.
Late 2017	Triton Malware Deployment	Triton malware was deployed against Schneider Electric Triconex Safety Instrumented Systems (SIS).
Late 2017	Operational Disruption	Abnormal controller behavior caused safety system failures and triggered operational shutdowns.
December 2017	Detection & Public Disclosure	The attack was investigated by cybersecurity researchers and publicly disclosed as one of the most advanced industrial cyberattacks.
2018 Onwards	Global Investigation &	Security organizations and

	Defense Improvements	industrial sectors improved monitoring, ICS security, and threat detection capabilities after the incident.
--	----------------------	---

Objectives of the Attack

- Gain unauthorized access to enterprise and industrial networks
- Steal credentials and sensitive operational data
- Maintain long-term persistence within compromised systems
- Perform lateral movement across internal environments
- Manipulate targeted industrial safety systems
- Evade security monitoring and detection mechanisms
- Establish remote command-and-control communication
- Potentially disrupt industrial operations and critical infrastructure systems

Targeted Sectors / Countries

Targeted Sectors : -

- Petrochemical Industry

- Industrial Control Systems (ICS)
- Operational Technology (OT) Environments
- Critical Infrastructure Organizations
- Energy and Manufacturing Sectors

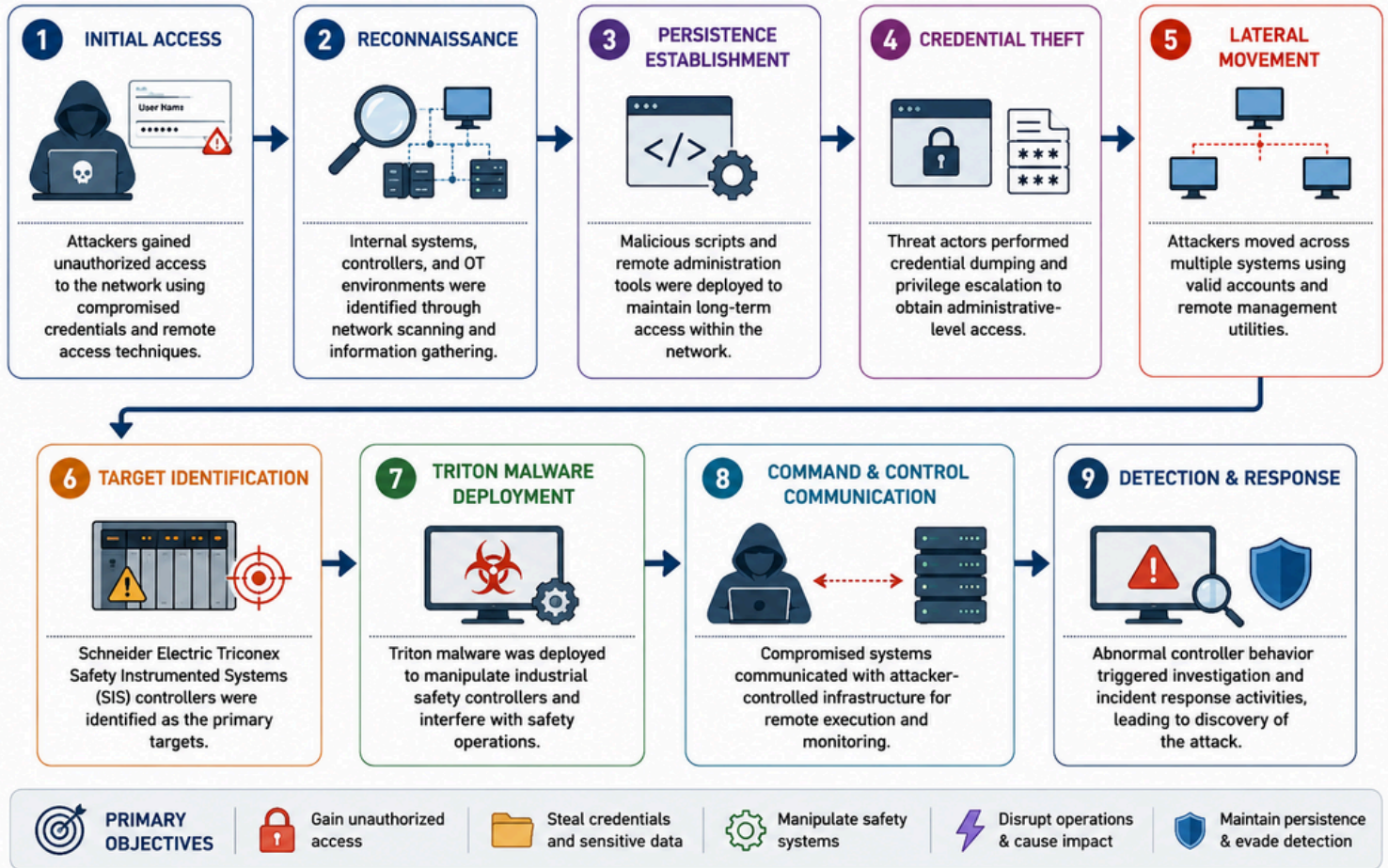
Targeted Countries : -

- Saudi Arabia (Primary confirmed target)
- Potential interest in global industrial and critical infrastructure environments due to the nature of the attack techniques used.

Attack flow

The C0032 campaign followed a multi-stage attack process involving initial access, reconnaissance, persistence, credential theft, and lateral movement across industrial networks. After identifying targeted Safety Instrumented Systems (SIS), the attackers deployed Triton malware to manipulate industrial safety controllers and maintain remote command-and-control communication until the attack was detected.

C0032 CAMPAIGN – ATTACK FLOW



MITRE ATT&CK Techniques Used

The C0032 campaign used multiple MITRE ATT&CK techniques to gain unauthorized access, maintain persistence, steal credentials, and move laterally across industrial networks. The attackers leveraged PowerShell scripts, remote services, credential dumping, and command-and-control communication to compromise Safety Instrumented Systems (SIS) and maintain stealth within the environment.

Key Techniques

- T1078 – Valid Accounts
- T1059 – Command and Scripting Interpreter

- T1003 – OS Credential Dumping
- T1021 – Remote Services
- T1547 – Boot or Logon Autostart Execution
- T1071 – Application Layer Protocol
- T1046 – Network Service Scanning

Technique ID	Technique Name	Purpose
T1078	Valid Accounts	Used compromised credentials for unauthorized access
T1059	Command and Scripting Interpreter	Executed malicious PowerShell and command scripts
T1003	OS Credential Dumping	Stole credentials from compromised systems
T1021	Remote Services	Enabled lateral movement using RDP/SMB/SSH
T1547	Boot or Logon Autostart Execution	Maintained persistence within systems
T1082	System Information Discovery	Gathered information about target systems
T1046	Network Service Scanning	Identified devices and services in the network
T1105	Ingress Tool Transfer	Transferred malicious

		tools and payloads
T1071	Application Layer Protocol	Used command-and-control communication
T1485	Data Destruction / Impact	Attempted disruption of industrial operations

Real-World Incident

In 2017, the Triton cyberattack targeted a petrochemical facility in Saudi Arabia by compromising Schneider Electric Triconex Safety Instrumented Systems (SIS). The attackers deployed Triton malware to manipulate industrial safety controllers responsible for preventing hazardous operational failures.

The attack caused abnormal controller behavior and operational shutdowns, leading to the discovery of the intrusion. This incident is considered one of the most advanced industrial cyberattacks because it directly targeted critical safety systems designed to protect infrastructure and human life.

Detection Opportunities

- Monitor abnormal remote access activity and unauthorized login attempts
- Detect unusual PowerShell and command-line execution within industrial environments.

- Identify credential dumping and privilege escalation behavior
- Monitor lateral movement using RDP, SMB, and remote administration tools
- Detect unauthorized communication with external command-and-control servers
- Monitor changes or abnormal activity in Safety Instrumented Systems (SIS) controllers
- Analyze network traffic for suspicious scanning and reconnaissance activity
- Use SIEM and EDR solutions to identify persistence mechanisms and malware execution
- Implement continuous monitoring in ICS and OT environments for operational anomalies

Defensive Recommendations

DEFENSIVE RECOMMENDATIONS – C0032 CAMPAIGN			
#	SECURITY AREA	DEFENSIVE RECOMMENDATION	PURPOSE
01	 ACCESS SECURITY	Implement Multi-Factor Authentication (MFA) Enforce MFA for all remote access, privileged accounts and critical systems.	 Prevent unauthorized access and account compromise.
02	 PRIVILEGE MANAGEMENT	Enforce Least Privilege Access Control Grant users only the minimum access necessary for their roles and regularly review permissions.	 Limit attacker privileges and reduce potential impact.
03	 POWERSHELL SECURITY	Restrict and Monitor PowerShell and Command-Line Execution Allow only required scripts, enable logging and alert on suspicious or abnormal command execution.	 Detect malicious scripts and unauthorized command activity.
04	 NETWORK SEGMENTATION	Segment IT and OT/ICS Environments Implement proper network segmentation and access controls between enterprise IT and industrial networks.	 Reduce lateral movement and contain threats within segments.
05	 ICS/OT PROTECTION	Continuously Monitor SIS and Industrial Controllers Monitor Safety Instrumented Systems (SIS) and other industrial controllers for abnormal changes or behavior.	 Detect abnormal activities that could disrupt industrial operations.
06	 ENDPOINT SECURITY	Deploy SIEM and EDR Solutions Use advanced monitoring to detect malware, persistence mechanisms and other malicious behaviors in real time.	 Improve visibility and enable faster threat detection and response.
07	 PATCH MANAGEMENT	Regularly Update and Patch Systems Keep operating systems, industrial software, firmware and network devices up to date.	 Reduce vulnerabilities exploited by attackers.
08	 REMOTE ACCESS MONITORING	Monitor Lateral Movement Activities Monitor RDP, SMB, WMI and other remote administration tools for suspicious activity.	 Detect attacker movement across the network early.
09	 SECURITY ASSESSMENT	Conduct Regular Security Audits Perform vulnerability assessments, penetration testing and configuration reviews in ICS/OT environments.	 Identify security gaps and strengthen overall security posture.
10	 INCIDENT RECOVERY	Maintain Backups and Response Plans Keep offline backups of critical systems and maintain an incident response plan for quick recovery.	 Ensure business continuity and quick recovery from security incidents.

- Strong access security using MFA
- Detection of malicious scripts and PowerShell activity
- Protection against credential theft and lateral movement
- Network segmentation for IT and OT/ICS security
- Continuous monitoring of industrial controllers and SIS systems
- Real-time threat detection using SIEM and EDR
- Regular patching and vulnerability management
- Backup and incident response readiness for critical systems

MITRE ATT&CK RESEARCH

SOFTWARE RESEARCH REPORT

Software Range S0320 – S0329 | Android Mobile Malware

Intern ID	2132
Assigned Campaign	Campaign No. 32 (C0032)
Software Range	S0320 – S0329 (10 Entries)
Platform Domain	MITRE ATT&CK Mobile

S0320 - DroidJack

1. Name & Type

- Name: DroidJack
- Type: Android Remote Access Trojan (RAT)
- MITRE ID: S0320

2. Description

DroidJack is an Android Remote Access Trojan (RAT) that allows attackers to remotely control infected mobile devices. It can steal data, monitor activity, record calls, capture messages, and access device functions without the user's knowledge.

3. Threat Actor Association

- Used by cybercriminals and spyware operators
- Common in mobile surveillance and espionage attacks

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Command Execution (T1059)	Execute remote commands
Input Capture (T1056)	Capture user activity
Audio Capture (T1123)	Record microphone audio
Screen Capture (T1113)	Capture screenshots
Data Exfiltration (T1041)	Steal mobile data

6. Execution Method

Delivered through:

- Fake Android applications
- Malicious APK files
- Third-party app stores
- Phishing links

7. Persistence Techniques

- Starts automatically after reboot
- Background services
- Hidden app icons
- Accessibility service abuse

8. Privilege Escalation

May:

- Abuse Android permissions
- Exploit device vulnerabilities
- Request administrator privileges

9. Defense Evasion

- Hidden application icons
- Obfuscated code
- Encrypted communication
- Runs silently in background

10. Credential Access

Targets:

- Saved passwords
- SMS messages
- Banking credentials
- Login information

11. Discovery Techniques

Collects:

- Device information
- Contact lists
- Installed applications
- Network information
- GPS location

12. Lateral Movement

Limited lateral movement, but attackers may:

- Send malicious links through infected devices

- Spread infected APK files

13. Command & Control (C2)

Communicates using:

- HTTP/HTTPS traffic
- Remote control servers
- Encrypted mobile communication

14. Exfiltration

Can steal:

- SMS messages
- Photos and videos
- Contacts
- Call logs
- Location data
- Credentials

15. Impact Analysis

Effects:

- Mobile surveillance
- Data theft
- Financial fraud
- Privacy compromise
- Unauthorized remote control

16. Indicators of Compromise (IOCs)

- Rapid battery drain
- Unusual background activity
- Unknown apps installed
- Increased mobile data usage
- Suspicious permissions

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor unusual permissions
- Analyze network traffic
- Check background services

Mitigation

- Install apps only from trusted stores
- Avoid unknown APK files
- Update Android devices
- Review app permissions
- Enable mobile security software

18. References

- MITRE ATT&CK
- Android security reports
- Mobile malware analysis research
- Threat intelligence documentation

S0321-HummingWhale

1. Name & Type

- Name: HummingWhale
- Type: Android Malware / Ad Fraud Malware
- MITRE ID: S0321

2. Description

HummingWhale is an Android malware family used mainly for ad fraud and malicious app activities. It creates fake virtual devices,

installs fraudulent applications, and generates fake advertisement revenue for attackers.

3. Threat Actor Association

- Associated with cybercriminal ad fraud operations
- Used for large-scale mobile advertising fraud campaigns

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
User Execution (T1204)	Victim installs malicious app
Virtualization/Sandbox Evasion	Hide malicious behavior
Command Execution (T1059)	Execute malicious actions
Data Exfiltration (T1041)	Transfer collected data
Obfuscated Files (T1027)	Hide malware code

6. Execution Method

Delivered through:

- Malicious Android apps
- Third-party app stores
- Fake applications
- Trojanized apps

7. Persistence Techniques

- Background services
- Auto-start on reboot
- Hidden processes
- Reinstallation mechanisms

8. Privilege Escalation

May:

- Abuse Android permissions
- Request device administrator access
- Exploit Android vulnerabilities

9. Defense Evasion

- Uses virtualization techniques
- Obfuscated code
- Encrypted communications
- Sandbox detection

10. Credential Access

Possible targets:

- User account data
- Device identifiers
- Advertising information
- Login details

11. Discovery Techniques

Collects:

- Device information
- Installed apps
- Network details

- User activity data

12. Lateral Movement

Limited lateral movement, but malware may:

- Spread through malicious app downloads
- Install additional applications

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted communication channels
- Remote configuration updates

14. Exfiltration

Can steal:

- Device information
- Advertising IDs
- User data
- Usage statistics

15. Impact Analysis

Effects:

- Mobile ad fraud
- Device slowdown
- Increased data usage
- Battery drain
- Privacy compromise

16. Indicators of Compromise (IOCs)

- Excessive advertisements

- Unknown applications installed
- High battery consumption
- Unusual network traffic
- Device performance issues

17. Detection & Mitigation

Detection

- Mobile antivirus solutions
- Network traffic monitoring
- Analyze installed applications
- Monitor abnormal background activity

Mitigation

- Install apps from trusted stores only
- Avoid suspicious APK files
- Keep Android updated
- Review app permissions
- Use mobile security software

18. References

- MITRE ATT&CK
- Android malware research reports
- Mobile threat intelligence reports
- Security analysis documentation

S0322 - HummingBad

1. Name & Type

- Name: HummingBad
- Type: Android Malware / Rooting Malware

- MITRE ID: S0322

2. Description

HummingBad is a powerful Android malware family that infects mobile devices to gain root access, install fraudulent apps, display advertisements, and generate illegal advertising revenue. It can fully compromise Android devices.

3. Threat Actor Association

- Associated with the Chinese advertising company Yingmob
- Used in large-scale mobile ad fraud campaigns

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Exploitation for Privilege Escalation (T1068)	Gain root access
User Execution (T1204)	Install malicious apps
Command Execution (T1059)	Execute malware commands
Obfuscated Files (T1027)	Hide malicious code
Data Exfiltration (T1041)	Steal device information

6. Execution Method

Delivered through:

- Malicious applications
- Third-party app stores
- Fake app downloads
- Trojanized apps

After installation, it attempts to root the device.

7. Persistence Techniques

- Root-level persistence
- Background services
- Auto-start on reboot
- Hidden applications

8. Privilege Escalation

- Exploits Android vulnerabilities
- Gains root privileges on infected devices
- Obtains full system control

9. Defense Evasion

- Obfuscated code
- Hidden background activity
- Encrypted communications
- Rootkit-like behavior

10. Credential Access

Possible targets:

- Google account data
- Device identifiers
- Login credentials
- User information

11. Discovery Techniques

Collects:

- Device information
- Installed applications
- Network configuration
- User activity data

12. Lateral Movement

Limited direct lateral movement, but malware may:

- Install additional malicious apps
- Spread through infected app downloads

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted communication channels
- Remote update mechanisms

14. Exfiltration

Can steal:

- Device information
- Advertising identifiers
- User data
- Account details

15. Impact Analysis

Effects:

- Full device compromise
- Ad fraud

- Unauthorized app installation
- Privacy violation
- Battery and performance issues

16. Indicators of Compromise (IOCs)

- Unknown apps installed
- Excessive advertisements
- Rapid battery drain
- High network usage
- Device overheating

17. Detection & Mitigation

Detection

- Mobile antivirus solutions
- Monitor rooting activity
- Analyze suspicious apps
- Monitor abnormal network traffic

Mitigation

- Install apps only from trusted stores
- Avoid unknown APK files
- Keep Android updated
- Remove rooted malicious apps
- Use mobile security software

18. References

- MITRE ATT&CK
- Check Point Research
- Android malware reports
- Mobile security documentation

1. Name & Type

- Name: Charger
- Type: Android Banking Trojan / Ransomware Malware
- MITRE ID: S0323

2. Description

Charger is an Android malware that combines banking trojan and ransomware capabilities. It steals user credentials, intercepts SMS messages, and can lock devices to demand ransom payments.

3. Threat Actor Association

- Used by cybercriminal groups
- Associated with financial fraud and mobile banking attacks

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Input Capture (T1056)	Steal user credentials
Screen Overlay	Fake banking login screens
SMS Collection	Intercept verification codes
Data Exfiltration (T1041)	Transfer stolen information
Device Locking	Ransomware functionality

6. Execution Method

Delivered through:

- Fake Android applications
- Third-party app stores
- Malicious APK files
- Phishing links

7. Persistence Techniques

- Auto-start on reboot
- Background services
- Device administrator abuse
- Hidden application icons

8. Privilege Escalation

May:

- Request administrator privileges
- Abuse Android permissions
- Exploit device vulnerabilities

9. Defense Evasion

- Obfuscated code
- Hidden app icons
- Encrypted communication
- Silent background execution

10. Credential Access

Targets:

- Banking credentials
- SMS OTP codes
- Login information

- Credit card details

11. Discovery Techniques

Collects:

- Device information
- Installed banking apps
- Contact lists
- SMS messages

12. Lateral Movement

Limited lateral movement, but may:

- Spread malicious APK links
- Send phishing SMS messages

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted communication channels
- Remote command servers

14. Exfiltration

Can steal:

- Banking data
- SMS messages
- Contacts
- Credentials
- Personal information

15. Impact Analysis

Effects:

- Financial theft
- Mobile ransomware attacks
- Credential compromise
- Privacy violation
- Unauthorized device control

16. Indicators of Compromise (IOCs)

- Fake banking screens
- Suspicious device administrator requests
- High battery/data usage
- Unknown apps installed
- Locked device behavior

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor suspicious permissions
- Analyze unusual app behavior
- Network traffic monitoring

Mitigation

- Install apps only from trusted stores
- Avoid suspicious APK files
- Do not grant unnecessary permissions
- Keep Android updated
- Use mobile security software

18. References

- MITRE ATT&CK
 - Android malware research reports
 - Mobile banking trojan analysis
 - Threat intelligence documentation
-

S0324 - SpyDealer

1. Name & Type

- Name: SpyDealer
- Type: Android Spyware / Remote Access Trojan (RAT)
- MITRE ID: S0324

2. Description

SpyDealer is an advanced Android spyware malware designed to steal sensitive information from infected mobile devices. It can monitor messages, record calls, capture screenshots, access contacts, and remotely control device functions.

3. Threat Actor Association

- Associated with cyber espionage operations
- Used in surveillance and targeted attacks

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Audio Capture (T1123)	Record calls and microphone audio
Screen Capture (T1113)	Capture screenshots
Input Capture (T1056)	Steal user input

Data Exfiltration (T1041)	Transfer stolen data
Command Execution (T1059)	Execute remote commands

6. Execution Method

Delivered through:

- Malicious Android applications
- Trojanized apps
- Third-party app stores
- Phishing links

7. Persistence Techniques

- Auto-start on reboot
- Background services
- Hidden application icons
- Accessibility service abuse

8. Privilege Escalation

May:

- Exploit Android vulnerabilities
- Abuse device permissions
- Request administrator privileges

9. Defense Evasion

- Obfuscated code
- Encrypted communication
- Hidden background execution
- Anti-analysis techniques

10. Credential Access

Targets:

- Login credentials
- SMS messages
- Banking information
- Social media accounts

11. Discovery Techniques

Collects:

- Device information
- Contacts
- Installed apps
- GPS location
- Network details

12. Lateral Movement

Limited direct lateral movement, but may:

- Spread through malicious APK files
- Send phishing links/messages

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted remote connections
- Remote control commands

14. Exfiltration

Can steal:

- SMS messages
- Call logs
- Photos/videos

- Contacts
- Credentials
- Location data

15. Impact Analysis

Effects:

- Mobile surveillance
- Privacy compromise
- Financial theft
- Credential theft
- Remote device control

16. Indicators of Compromise (IOCs)

- Rapid battery drain
- Unusual background activity
- Unknown applications installed
- High mobile data usage
- Suspicious permissions requests

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor suspicious app permissions
- Analyze unusual network traffic
- Check background services

Mitigation

- Install apps only from trusted stores
- Avoid unknown APK files
- Keep Android updated
- Review app permissions carefully
- Use mobile security software

18. References

- MITRE ATT&CK
 - Android spyware research reports
 - Mobile malware analysis documentation
 - Threat intelligence reports
-

S0325 -Judy

1. Name & Type

- Name: Judy
- Type: Android Adware Malware
- MITRE ID: S0325

2. Description

Judy is a large-scale Android adware malware family that infects devices through malicious applications. It generates fraudulent advertisement clicks in the background to produce illegal advertising revenue for attackers.

3. Threat Actor Association

- Linked to mobile ad fraud operations
- Distributed through infected Android applications on app stores

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
User Execution (T1204)	Victim installs malicious app
Command Execution (T1059)	Run malicious background activity
Obfuscated Files (T1027)	Hide malware code
Web Protocols	Generate fake advertisement clicks
Data Exfiltration (T1041)	Send device information

6. Execution Method

Delivered through:

- Malicious Android applications
- Fake utility or gaming apps
- Third-party app stores
- Infected apps uploaded to Google Play Store

7. Persistence Techniques

- Background services
- Auto-start after reboot
- Hidden app activity
- Silent advertisement processes

8. Privilege Escalation

Usually does not require high privileges, but may:

- Abuse Android permissions

- Request excessive app permissions

9. Defense Evasion

- Obfuscated code
- Hidden background execution
- Uses legitimate-looking applications
- Delayed malicious activity

10. Credential Access

Not mainly focused on credential theft, but may collect:

- Device identifiers
- User information
- App usage data

11. Discovery Techniques

Collects:

- Device information
- Network details
- Installed applications
- Advertising identifiers

12. Lateral Movement

Limited lateral movement, but may:

- Spread through shared app links
- Install additional malicious apps

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Advertisement-related network requests
- Remote configuration updates

14. Exfiltration

Can send:

- Device information
- User activity data
- Advertising IDs
- Usage statistics

15. Impact Analysis

Effects:

- Excessive advertisements
- Battery drain
- Increased mobile data usage
- Device slowdown
- Privacy concerns

16. Indicators of Compromise (IOCs)

- Excessive popup advertisements
- Unknown applications installed
- High battery consumption
- Increased network traffic
- Slow device performance

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Analyze suspicious apps
- Monitor abnormal network traffic

- Check excessive advertisement activity

Mitigation

- Install apps only from trusted stores
- Avoid suspicious APK files
- Remove unknown applications
- Keep Android updated
- Use mobile security software

18. References

- MITRE ATT&CK
 - Check Point Research
 - Android malware analysis reports
 - Mobile security documentation
-

S0326 -RedDrop

1. Name & Type

- Name: RedDrop
- Type: Android Spyware / Trojan
- MITRE ID: S0326

2. Description

RedDrop is an Android spyware malware family that steals sensitive user data from infected mobile devices. It can collect audio recordings, contacts, photos, SMS messages, and device information while secretly communicating with attacker-controlled servers.

3. Threat Actor Association

- Associated with mobile cybercriminal operations

- Used for surveillance and data theft activities

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Audio Capture (T1123)	Record microphone audio
Screen/Data Collection	Steal user information
Command Execution (T1059)	Execute malware commands
Data Exfiltration (T1041)	Send stolen data
Obfuscated Files (T1027)	Hide malicious code

6. Execution Method

Delivered through:

- Fake Android applications
- Trojanized apps
- Third-party app stores
- Phishing links

7. Persistence Techniques

- Auto-start on reboot
- Background services
- Hidden application behavior

- Silent execution in background

8. Privilege Escalation

May:

- Abuse Android permissions
- Request excessive permissions
- Exploit Android vulnerabilities

9. Defense Evasion

- Obfuscated code
- Hidden app functionality
- Encrypted communication
- Delayed malicious actions

10. Credential Access

Possible targets:

- Login credentials
- SMS messages
- User account information
- Personal data

11. Discovery Techniques

Collects:

- Device information
- Contacts
- Photos and media files
- GPS/location data
- Installed apps

12. Lateral Movement

Limited lateral movement, but malware may:

- Spread through malicious APK files
- Send phishing messages or links

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted remote connections
- Remote configuration updates

14. Exfiltration

Can steal:

- Audio recordings
- SMS messages
- Contacts
- Photos/videos
- Device information
- Credentials

15. Impact Analysis

Effects:

- Mobile surveillance
- Privacy compromise
- Data theft
- Financial fraud risks
- Unauthorized device monitoring

16. Indicators of Compromise (IOCs)

- High battery consumption
- Excessive data usage
- Unknown apps installed
- Suspicious microphone activity

- Slow device performance

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor unusual permissions
- Analyze suspicious network traffic
- Check abnormal background activity

Mitigation

- Install apps only from trusted stores
- Avoid unknown APK files
- Keep Android updated
- Review app permissions carefully
- Use mobile security software

18. References

- MITRE ATT&CK
- Android spyware research reports
- Mobile malware analysis documentation
- Threat intelligence reports

S0327 -Skygofree

1. Name & Type

- Name: Skygofree
- Type: Android Spyware / Remote Access Trojan (RAT)
- MITRE ID: S0327

2. Description

Skygofree is a sophisticated Android spyware malware capable of monitoring users, stealing data, recording audio, tracking location, and remotely controlling infected devices. It is known for advanced surveillance capabilities.

3. Threat Actor Association

- Associated with mobile espionage operations
- Believed to be used in targeted surveillance attacks

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Audio Capture (T1123)	Record microphone audio
Screen Capture (T1113)	Capture screenshots
Input Capture (T1056)	Steal user input
Data Exfiltration (T1041)	Transfer stolen data
Command Execution (T1059)	Execute remote commands

6. Execution Method

Delivered through:

- Fake mobile applications
- Malicious APK files

- Fake update pages
- Phishing links

7. Persistence Techniques

- Auto-start after reboot
- Background services
- Accessibility service abuse
- Hidden application behavior

8. Privilege Escalation

May:

- Exploit Android vulnerabilities
- Abuse administrator permissions
- Request excessive device permissions

9. Defense Evasion

- Obfuscated code
- Encrypted communications
- Hidden background activity
- Anti-analysis techniques

10. Credential Access

Targets:

- Login credentials
- Banking information
- SMS OTP codes
- Social media accounts

11. Discovery Techniques

Collects:

- Device information

- Contacts
- GPS location
- Installed applications
- Network information

12. Lateral Movement

Limited direct lateral movement, but malware may:

- Spread malicious APK files
- Send phishing messages/links

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted remote channels
- Remote control commands

14. Exfiltration

Can steal:

- SMS messages
- Audio recordings
- Photos/videos
- Contacts
- Credentials
- Location data

15. Impact Analysis

Effects:

- Mobile surveillance
- Privacy compromise
- Credential theft

- Financial fraud
- Remote device control

16. Indicators of Compromise (IOCs)

- Rapid battery drain
- Increased data usage
- Suspicious permissions requests
- Unknown apps installed
- Microphone activating unexpectedly

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor suspicious app permissions
- Analyze unusual network traffic
- Check abnormal background services

Mitigation

- Install apps only from trusted stores
- Avoid unknown APK files
- Keep Android updated
- Review app permissions carefully
- Use mobile security software

18. References

- MITRE ATT&CK
- Android spyware research reports
- Mobile malware analysis documentation
- Threat intelligence reports

S0328-Stealth Mango

1. Name & Type

- Name: Stealth Mango
- Type: Android Spyware / Surveillance Malware
- MITRE ID: S0328

2. Description

Stealth Mango is an Android spyware malware used for surveillance and information theft. It can monitor user activity, collect device information, steal files, and communicate with remote attacker-controlled servers.

3. Threat Actor Association

- Associated with targeted espionage operations
- Used in surveillance campaigns against mobile users

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Audio Capture (T1123)	Record microphone audio
Screen Capture (T1113)	Capture screenshots
Input Capture (T1056)	Steal user input
Command Execution (T1059)	Execute remote commands
Data Exfiltration (T1041)	Transfer stolen information

6. Execution Method

Delivered through:

- Malicious Android applications
- Trojanized APK files
- Phishing links
- Fake update applications

7. Persistence Techniques

- Background services
- Auto-start after reboot
- Hidden app components
- Accessibility service abuse

8. Privilege Escalation

May:

- Exploit Android vulnerabilities
- Abuse administrator permissions
- Request excessive permissions

9. Defense Evasion

- Obfuscated code
- Encrypted communications
- Hidden background execution
- Anti-analysis techniques

10. Credential Access

Targets:

- Login credentials
- SMS messages
- Banking information

- Social media accounts

11. Discovery Techniques

Collects:

- Device information
- Contacts
- Installed apps
- GPS location
- Network configuration

12. Lateral Movement

Limited lateral movement, but malware may:

- Spread malicious APK files
- Send phishing links/messages

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted communication channels
- Remote control commands

14. Exfiltration

Can steal:

- Contacts
- SMS messages
- Photos/videos
- Credentials
- Audio recordings
- Location data

15. Impact Analysis

Effects:

- Mobile surveillance
- Privacy compromise
- Credential theft
- Financial fraud risks
- Remote monitoring of victims

16. Indicators of Compromise (IOCs)

- High battery consumption
- Increased mobile data usage
- Unknown apps installed
- Suspicious permission requests
- Slow device performance

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor suspicious permissions
- Analyze abnormal network traffic
- Check background processes

Mitigation

- Install apps only from trusted stores
- Avoid suspicious APK files
- Keep Android updated
- Review app permissions carefully
- Use mobile security software

18. References

- MITRE ATT&CK

- Android spyware research reports
- Mobile malware analysis documentation
- Threat intelligence reports

S0329- Tangelo

1. Name & Type

- Name: Tangelo
- Type: Android Spyware / Surveillance Malware
- MITRE ID: S0329

2. Description

Tangelo is an Android spyware malware used for surveillance and information theft. It can collect user data, monitor device activity, steal files, and communicate with remote attacker-controlled servers.

3. Threat Actor Association

- Associated with targeted mobile surveillance campaigns
- Used in espionage and cyber monitoring operations

4. Supported Platforms

- Android smartphones
- Android tablets

5. ATT&CK Techniques Used

Technique	Purpose
Audio Capture (T1123)	Record microphone audio

Screen Capture (T1113)	Capture screenshots
Input Capture (T1056)	Steal user input
Command Execution (T1059)	Execute attacker commands
Data Exfiltration (T1041)	Transfer stolen information

6. Execution Method

Delivered through:

- Malicious Android applications
- Trojanized APK files
- Phishing links
- Fake software updates

7. Persistence Techniques

- Background services
- Auto-start after reboot
- Hidden app components
- Accessibility service abuse

8. Privilege Escalation

May:

- Exploit Android vulnerabilities
- Abuse administrator permissions
- Request excessive permissions

9. Defense Evasion

- Obfuscated code
- Encrypted communications
- Hidden background activity

- Anti-analysis techniques

10. Credential Access

Targets:

- Login credentials
- SMS messages
- Banking information
- Social media accounts

11. Discovery Techniques

Collects:

- Device information
- Contacts
- Installed applications
- GPS location
- Network configuration

12. Lateral Movement

Limited lateral movement, but malware may:

- Spread through malicious APK files
- Send phishing links/messages

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS traffic
- Encrypted communication channels
- Remote command execution

14. Exfiltration

Can steal:

- Contacts
- SMS messages
- Photos/videos
- Credentials
- Audio recordings
- Location data

15. Impact Analysis

Effects:

- Mobile surveillance
- Privacy compromise
- Credential theft
- Financial fraud risks
- Unauthorized remote monitoring

16. Indicators of Compromise (IOCs)

- Rapid battery drain
- Increased data usage
- Suspicious app permissions
- Unknown apps installed
- Device slowdown

17. Detection & Mitigation

Detection

- Mobile antivirus tools
- Monitor suspicious permissions
- Analyze unusual network traffic
- Check abnormal background activity

Mitigation

- Install apps only from trusted stores
- Avoid unknown APK files

- Keep Android updated
- Review app permissions carefully
- Use mobile security software

18. References

- MITRE ATT&CK
- Android spyware research reports
- Mobile malware analysis documentation
- Threat intelligence reports

Summary & Conclusion

- This report analyzed the MITRE ATT&CK Campaign C0032 (Triton/TRISIS) and Android malware families S0320–S0329 to understand modern cyberattack techniques and threat actor behavior.
 - The Triton attack demonstrated how advanced attackers targeted industrial Safety Instrumented Systems (SIS), causing operational disruption and highlighting risks to critical infrastructure.
 - The malware analysis covered spyware, banking trojans, adware, ransomware, and surveillance malware using credential theft, persistence, remote access, privilege escalation, and encrypted command-and-control communication.
 - The research mapped real-world cyberattacks with MITRE ATT&CK techniques such as T1078, T1059, T1003, T1021, and T1071.
 - Key findings emphasize the importance of Cyber Threat Intelligence (CTI), SIEM/EDR monitoring, MFA, network segmentation, continuous monitoring, and mobile security practices.
 - Overall, the report demonstrates how proactive defense strategies and threat intelligence can help protect enterprise, industrial, and mobile environments from advanced cyber threats.
-